Sprint 6 Validation Report

Bluestock Fintech - Nifty 100 Financial Intelligence Platform

Generated 27 October 2026 | Python 3.14.3 / 3.11.9, FastAPI 0.115+, Pytest 9.1.1, Scikit-Learn 1.6+

Summary

Sprint 6 delivered KMeans clustering, a 16-endpoint FastAPI server, a 130-test pytest suite and the analyst documentation with acceptance checklist. All four modules are marked verified complete in the audit trace, and all 130 tests passed with zero failures.

Module	Deliverable	Status
10	KMeans clustering and statistical profiling	Verified complete
11	FastAPI REST server (16 endpoints)	Verified complete
12	Quality assurance and test suite	Verified complete (zero failures)
13	Documentation and acceptance checklist	Verified complete

Module 10 - KMeans Clustering and Statistical Profiling

Script: `src/analytics/clustering.py`

Algorithm: StandardScaler + KMeans(n_clusters=5, random_state=42)
Features: ROE, D/E, Revenue CAGR 5yr, FCF CAGR 5yr, OPM
Imputation: sector median, with fallback to global median

Outputs

- `output/cluster_labels.csv` - 100 companies assigned to 5 labeled clusters
- `reports/elbow_plot.png` - inertia curve for k=1..9, clear elbow at k=4..5
- `reports/correlation_heatmap.png` - correlation matrix across the 5 core features
- `output/outlier_report.csv` - outliers flagged at |z-score| > 3.0 against sector
- `output/portfolio_stats.csv` - P10-P90 and mean/median per cluster

Cluster	Profile
0	High Margin Leaders
1	High Quality / Low Debt
2	High Growth
3	Highly Leveraged
4	Mixed Operating Profile

Status: **Verified complete**

Module 11 - FastAPI REST Server (16 Endpoints)

Script: `src/api/main.py`

Engine: FastAPI with Uvicorn, CORS middleware enabled, X-Process-Time header

Specification documents: `docs/openapi.json` (OpenAPI 3.1.0, 16.3 KB); `docs/postman_collection.json` (Postman v2.1, 6.8 KB)

Verified endpoints (16 / 16)

#	Method	Path	Returns (all status 200)
1	GET	`/api/v1/health`	Uptime, DB row counts
2	GET	`/api/v1/companies`	92 companies
3	GET	`/api/v1/companies/{ticker}`	Profile; 404 on invalid
4	GET	`/api/v1/companies/{ticker}/pl`	10-14yr P&L history
5	GET	`/api/v1/companies/{ticker}/bs`	Balance sheet history
6	GET	`/api/v1/companies/{ticker}/cashflow`	CFO, CFI, CFF history
7	GET	`/api/v1/companies/{ticker}/ratios`	51 computed ratios
8	GET	`/api/v1/companies/{ticker}/tearsheet`	Binary PDF stream
9	GET	`/api/v1/screener`	Ranked multi-filter list
10	GET	`/api/v1/sectors`	Distinct sector list
11	GET	`/api/v1/sectors/{sector}/companies`	Companies in sector
12	GET	`/api/v1/peers/{group_name}`	11 peer groups
13	GET	`/api/v1/companies/{ticker}/peers/compare`	Radar comparisons
14	GET	`/api/v1/market-cap/{ticker}`	Valuation multiples
15	GET	`/api/v1/portfolio/stats`	Cluster distribution stats
16	GET	`/api/v1/companies/{ticker}/documents`	Annual report links

Status: **Verified complete**

Module 12 - Quality Assurance and Test Suite

Framework: pytest 9.1.1 + pytest-html. **Report:** `reports/pytest_report.html` (self-contained, 110 KB)

Measure	Result
Tests collected	130
Tests passed	130
Tests failed	0
Execution duration	6.30 seconds

Category	Tests	Coverage
ETL Normalisation	43	Year formatting, ticker case-normalisation, deduplication
Data Quality Rules	15	DQ01 through DQ16 bounds, integrity checks
KPI Formulas	24	ROE bounds, D/E edge cases, turnaround CAGR, ICR debt-free logic

Category	Tests	Coverage
Screener Engine	7	Preset filtering, ranking logic, composite score boundedness
Integration and API	41	All 16 routes, 404 validation, schema validity

Status: **Verified complete (zero failures)**

Module 13 - Documentation and Acceptance Checklist

File	Size	Contents
`docs/analyst_guide.pdf`	8.7 KB	10 sections covering Dashboard, Screener, Valuation, NLP, Tearsheets, Clustering, API
`docs/acceptance_checklist.pdf`	2.7 KB	17 formal audit checkpoints evaluated

Status: **Verified complete**

Observation

The clustering outputs cover 100 companies, while the API's companies endpoint and Sprint 5 reports cover 92. This is worth confirming (for example, whether 8 companies lack sufficient data for the API database) so the portfolio-wide figures are described consistently.